Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
1st Year 1st Semester Final Examination, 2024
MATH 1107: Calculus I
Marks: 60 # Duration: 3 hours

Md. Tawnat Hossen Nibir
BSSE 1625

Professionalism Excellence Respect

[Answer any five questions. When answering a question, please answer all the subsections of it at once]

1. a. If $\lim_{x\to 0}\frac{\sin 2x+a\sin x}{x^3}$ is finite, then find the value of a and the limiting value. 6

b. Give example of a function $f(x)$ such that $f'(x)$ is continuous but $f''(x)$ is not. Provide your explanation. 6

2. a. Show that $f(x) = \begin{cases} x\sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$ is continuous but not differentiable at x=0. 12

b. Find all lines that are simultaneously tangent to the graph of $y = x^2 + 1$ and to the graph of $y = -x^2 - 1$.

c. Compute the limit $\lim_{x\to\infty}[\ln x - \ln(1+x)]$.

3. a. Consider the function $f(x) = 5 + 12x - x^3$ on $[-5, 5]$, and Find: 12

i. the intervals on which f is increasing,
ii. the intervals on which f is decreasing,
iii. the open intervals on which f is concave up,
iv. the open intervals on which f is concave down,
v. all the inflection points,
vi. local maximum and local minimum, and
vii. absolute maximum and minimum.

4. a. Determine if the series $\sum\frac{1}{n\ln n}$ is convergent or divergent. 12

b. Determine if the series $\sum\frac{3^{1-2n}}{n^2-1}$ converges or diverges.

c. Determine if the series $\sum\frac{10^{3+2n}}{n^{n^2-1}}$ converges or diverges.

5. a. Find the area of the region enclosed by the curve $r = 1 + \cos\theta$.

b. Find the exact arc-length of the curve $8xy^2 = y^6 + 2$ from $y = 1$ to $y = 4$. 12

6. a. Determine the length of $\ln\sec x$ on $0 \le x \le \frac{\pi}{4}$.

b. Find the volume of the surface of revolution of the area under the curve $f(x) = \frac{1}{x^{\frac{3}{2}}}$ and above x-axis on $1 \le x \le \infty$ about the x-axis. 12

c. Find the surface area of the surface of revolution of the area under the curve $f(x) = \frac{1}{x}$ and above x-axis on $1 \le x \le \infty$ about the x-axis.

7. a. Use the Bisection method to approximate the value of $\sqrt{5}$ correct up to four decimal places. 12

b. Use the Newton Raphson method to approximate the value of $\sqrt{3}$ correct up to four decimal places.

c. Use the Fixed point iteration method to approximate the value of $\sqrt{7}$ correct up to four decimal places.

Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
1st Year 1st Semester Final Examination, 2024
CSE 1102 : Discrete Mathematics Md. Tawsat Hossen Nibir
Marks: 60 # Duration: 3 hours BSSE 1625

Professionalism Excellence Respect

[Answer any five questions. When answering a question, please answer all the subsections of it at once]

1. a. Express the follow statements using predicates and quantifiers. 3
"Every student in this class has studied calculus"
"Some student in this class has visited Mexico"
"Every student in this class has visited either Canada or Mexico"

b. Use a truth table to verify the first De Morgan law $\neg(p \wedge q) \equiv \neg p \vee \neg q$. 3

c. Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed." 4

d. What are Universal modus ponens and tollens rules? 2

2. a. Prove that if n is an integer and $3n + 2$ is odd, then n is odd. 3

b. Let A, B, and C be sets. Show that $A \cup (B \cap C) = (C \cup B) \cap A$. 3

c. Define fallacy of affirming the conclusion. Give an example. 2

d. Define proper subset using implication and quantifiers. What is the Cartesian product $A \times B \times C$, where $A = \{1, 2\}$, $B = \{2, 3\}$, and $C = \{1, 2, 3\}$? 4

3. a. Consider the following relations on $\{1, 2, 3, 4\}$: 6

$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$,
$R2 = \{(1, 1), (1, 2), (2, 1)\}$,
$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,
$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,
$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$,
$R6 = \{(3, 4)\}$.

Which of these relations are reflexive, symmetric, anti-symmetric, and transitive?

b. What is recurrence relation? Express Fibonacci sequence using recurrence relation. 2

c. Use mathematical Induction to prove that $2n < n!$ for every integer n with $n \geq 4$. (Note that this inequality is false for $n = 1$, 2, and 3.) 3

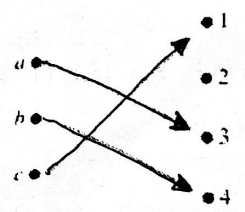
d. What is a well-ordered set? Give example. 1

4. a. Suppose that the relations $R1$ and $R2$ on a set A are represented by the matrices 4

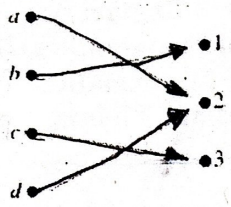
$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

What are the matrices representing $R1 \cup R2$ and $R1 \cap R2$?

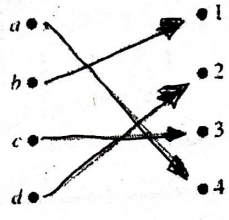
b. Identify following as one-to-one, onto, both one-to-one and onto, neither of any and not a function. Mention also whether the given mapping is not onto while it is one-to-one and vice-versa. 5



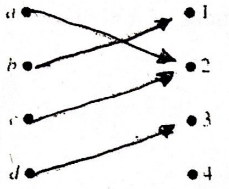
i.



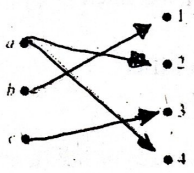
ii.



iii.

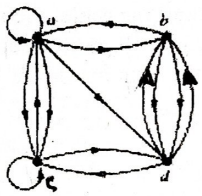


iv.



v.

c. Find the adjacency matrix of the given directed multigraph with respect to the vertices listed in alphabetic order. 3



5. a. How many bit strings are there of length six or less, not counting the empty string? 2

b. How many bit strings of length ten both begin and end with a 1? 2

c. How many license plates consisting of three letters followed by three digits contain no letter or digit twice? 3

d. Use binomial theorem to determine the coefficient of x^7 in $(1+x)^{11}$ 2

e. A coin is flipped eight times where each flip comes up either heads or tails. How many possible outcomes: 3

i. are there in total?

ii. contain exactly three heads?

iii. contain at least three heads?

a. Construct a precedence graph for the following program: 4

$S1: x := 0$
$S2: x := x + 1$
$S3: y := 2$
$S4: z := y$
$S5: x := x + 2$
$S6: y := x + z$
$S7: z := 4$

b. Suppose that a new company has five employees: Zaman, Ahmed, Siddique, Chowdhury, and Mahmud. Each employee will assume one of six responsibilities: planning, publicity, sales, marketing, development, and industry relations. Each employee is capable of doing one or more of these jobs: Zaman could do planning, sales, marketing, or industry relations; Ahmed could do planning or development; Siddique could do publicity, sales, or industry relations; Chowdhury could do planning, sales, or industry relations; and Mahmud could do planning, publicity, sales, or industry relations. 8

i. Model the capabilities of these employees using a partite graph.

ii. Find an assignment of responsibilities such that each employee is assigned one responsibility.

iii. Is the matching of responsibilities you found in part (b) a complete matching?

iv. Is it a maximum matching?

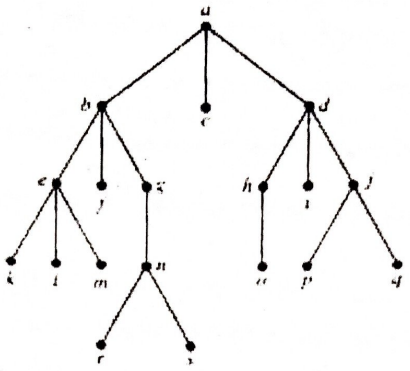
$n = mi + l$ $n = i + 1$

a. i. How many edges does a full binary tree with 1000 internal vertices have? **3**

ii. How many leaves does a full 3-ary tree with 100 vertices have? **+ 3**

Justify your answers.

b. Perform *Preorder traversal* of the following tree: **6**



Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
1st Year 1st Semester Final Examination, 2024
CSE 1101: Structured Programming
Marks: 60 # Duration: 3 hours

Md. Tawrat Hossen Nibin
BSSE 1625

Professionalism Excellence Respect

[Answer any five questions. When answering a question, please answer all the subsections of it at once]

1. a. Explain the advantages of using preprocessor directives such as **#define** and **#include**. How do these directives contribute to code organization and maintainability? 4

   b. What are the data type modifiers? How type conversions in expressions are performed in C? Provide an example code. 3

   c. Write a C program that reads an integer then finds the prime numbers within that number, after that checks if it can be represented as a sum of two prime numbers. 5

2. a. Explain how nested `if-else` statements work. Show where it is preferable over a `switch` statement. Write a code to identify whether a character is vowel or not using switch and its corresponding code using `if-else`. 4

   b. What are the significances of `break`, and `continue`? 2

   c. Write a C program that takes a year as input and determines if it is a **leap year** using a `switch` statement. Then determine if a given **date (dd/mm/yyyy) is valid**, considering leap years and varying days in months. 6

3. a. Explain the concept of **pointer arithmetic** with examples of increment and decrement operations on pointers. 3

   b. Show that pointers and arrays are interchangeable with example codes. 4

   c. Write a code which will take a string from the command line and use a recursive function to reverse the string using pointers. 5

4. a. Define what recursion is. Explain the two essential components of a recursive function. 3

   b. Write a recursive function to calculate the factorial of a given non-negative integer. Explain how your function works and ensures where to return after each function call. 5

   c. Discuss the importance of writing clean, maintainable code. What are some of the key principles of good code style? (e.g., naming conventions, comments, indentation, modularity). 4

5. a. Explain the difference between 'call by value' and 'call by reference' parameter passing methods in the context of functions. What are the implications of each method on the original variables passed to the function? Explain with appropriate examples. 6

   b. Write a C function called **findMax** that takes an integer array and its size as input and returns the largest element in the array. 6

6. a. What do you understand by `file` and `stream`? List all the common file read and write functions. 3

   b. How can we randomly access a file? Give an example code. 3

   c. Write a C program to read an **existing file** and count the number of **words and lines** in it. Then **copy the contents** of one file into another while replacing all occurrences of a given word with another word. 6

7. a. Define and differentiate between Structure and Union. How are they different than arrays? 4

b. What is bit-field? Write a structure where you can put a student's roll, batch and semester information in a single integer. Then store the information of 56 students of BSSE16, where you can retrieve information by searching any of the fields. 4

c. Take an input text file and encrypt each two consecutive characters by swapping these two and write the encrypted text to a new file. 4

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Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
1st Year 1st Semester Final Examination, 2024
STAT 1103: Probability and Statistics for Engineers I
Marks: 60 # Duration: 3 hours

Md. Tawnat Hossen Nibir
BSSE 1625

Professionalism Excellence Respect

[Answer any five questions. When answering a question, please answer all the subsections of it at once]

1. The following table shows the distribution of CGPA obtained from 120 third-year students of Software Engineering.

CGPA	Frequency
3.10	8
3.25	8
3.50	44
3.65	27
3.77	13
3.90	15
3.95	5

a. Calculate the sample mean and median of CGPA. Based on these two values, comment on the shape of the distribution. 4

b. Calculate the variance and standard deviation of CGPA. What is the advantage of standard deviation over variance as a measure of spread? 5

c. Calculate the coefficient of variation (CV) of CGPA. Describe the use of CV in statistical analysis. 3

2. a. What are the axioms of probability? Using these axioms, show that $P(A) = 1 - P(A^c)$. 4

b. In a study of children in grades 2 through 4, a researcher gave each student a reading test. When looking at the resulting data, the researcher noted a positive correlation between a student's reading test score and height. The researcher concluded that taller children read better because they can more easily see the blackboard. What do you think? 2

c. Two local factories that produce radios. Each radio produced at factory A is defective with probability 0.05, whereas each produced at factory B is defective with probability 0.01. Also assume $P(A) = P(B) = 0.5$. 6

i. What is the probability of purchasing a defective radio?
ii. What is the probability of purchasing two defective radios from factory A?
iii. Suppose you purchase two radios that were produced at the same factory, if the first radio that you check is defective, what is the conditional probability that the other one is also defective?

3. a. Define the moment generating function for a random variable. How can you obtain $E(X^3)$ from a moment generating function? 3

b. If $E(X) = 2$ and $E(X^2) = 8$, calculate $E[(2 + 4X)^2]$ and $E[X^2 + (X + 1)^2]$. 3

c. Suppose two random variable X and Y have the following joint density function 6

$$f(x, y) = x + y,\ 0 \le x, y \le 1.$$

Find the marginal densities of X, $E(Y)$, $V(X)$, and $P(X = 0.5)$.

4. a. At least half of an airplane's engines are required to function for it to operate. If each engine independently functions with probability p, for what values of p is a 4-engine plane more likely to operate than a 2-engine plane? 3

b. Derive the moment generating function of X_1, which follows a Poisson distribution with the parameter λ. Also, derive the moment generating function of $Y = X_1 + X_2$, where X_1 and X_2 are independent Poisson random variables with parameter λ. What is the distribution of Y? 5

c. The annual rainfall (in inches) in a certain region is normally distributed with $\mu = 40$, $\sigma = 4$. What is the probability that in 2 of the next 4 years, the rainfall will exceed 50 inches? Assume that the rainfalls in different years are independent. 4

5. a. Define sampling distribution. How does t-statistic differ from a standard normal distribution? 3

b. What is the central limit theorem? Let $X_1, \ldots, X_n$ be a random sample from an exponential distribution with parameter λ. Using the central limit theorem, obtain the distribution of the sample mean $\bar{X} = \sum_{i=1}^{n} X_i / n$. 5

c. A tobacco company claims that the amount of nicotine in its cigarettes is a random variable with a mean of 2.2 mg and a standard deviation of 0.3 mg. However, the sample mean nicotine content of 100 randomly chosen cigarettes was 3.1 mg. What is the approximate probability that the sample mean would have been as high or higher than 3.1 if the company's claims were valid? 4

6. a. What is the main difference between inferential and descriptive statistics? 2

b. What do you understand by estimation in statistical inference? Describe the maximum likelihood method of estimation. Can a maximum likelihood estimator of a distribution with a single parameter be obtained using a graphical method? If so, how? 5

c. Let $X_1, \ldots, X_n$ be a random sample from N $(80, \sigma^2)$. Obtain the expression of the maximum likelihood estimator of σ^2. 5

a. Discuss the concept of maximum likelihood estimation. 3

b. Let $X_1, \ldots, X_n$ be a random sample from a distribution with the probability mass function $p(x; \Theta) = \frac{\Theta^x e^{-\Theta}}{x!}$. Derive the expressions of the log-likelihood function and the maximum likelihood estimator $\hat{\Theta}$. 6

c. Show that $\hat{\Theta}$ is unbiased, i.e. $E(\hat{\Theta}) = \Theta$ and $Var(\hat{\Theta}) = \Theta^2/n$ 3

$\lambda e^{-\lambda x}$

$e^{tx} \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

$e^{-\lambda} \cdot \int \frac{e^{tx} \lambda^x}{x!}$

$\int \frac{(e^t \lambda)^x}{x!}$

$e^{-\lambda} \frac{\lambda \cdot}{0!}$

$\frac{1}{n} \quad -\frac{1}{n^2}$

Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
1st Year 1st Semester Final Examination, 2024
SE 1106: Introduction to Software Engineering
Marks: 60 # Duration: 3 hours

Md. Tawnat Hossen Nihal
BSSE 1625

Hello World

Professionalism Excellence Respect

[Answer any five questions. When answering a question, please answer all the subsections of it at once]

1. a. What are the various categories of software? Give examples of each. 4

   b. Suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems. Explain your answer according to the type of system being developed: 4
   - A system to control antilock braking in a car W
   - A virtual reality system to support software maintenance I
   - A university accounting system that replaces an existing system UBSE
   - An interactive travel planning system that helps users plan journeys with the lowest environmental impact UBSE I

   c. Describe the characteristics of software in contrast to characteristics of hardware. 4

2. a. What is Scrum? Describe the overall flow of the Scrum process with appropriate diagram. 8

   b. Discuss the advantages and disadvantages of Scrum. 4

3. a. What are the different types of software teams and their activities in a particular scenario where people will be working for k years and perform m different functional tasks. 7

   b. Describe agility for software projects. When would you recommend against the use of an agile method for developing a software system? Justify your answer. 5

4. a. What is software measurement? Write and explain the different types of software measurement. 5

   b. Write the names of different metrics used in software engineering? Explain in detail the Function Oriented metrics. 7

5. a. Define Software Project Management? Write down different causes which may affect the success of a project. 5

   b. Describe the common approaches for developing a software project. 7

6. a. What is software engineering? List the goals of software engineering. 4

   b. Suggest four reasons why the productivity rate of programmers working as a pair might be more than half that of two programmers working individually. 4

   c. Explain the basic principle of Prototyping with an example. 4

7. a. Write the objectives and principles of software testing. 6

   b. Define software testability. What are its characteristics? Write the attributes of good software test. 6

Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
1st Year 1st Semester Final Examination, 2024
GE 1105: Introduction to Sociology
Marks: 60 # Duration: 3 hours

Md. Tawnat Hossen Nibir
BSSE 1625

Professionalism Excellence Respect

[Answer any five questions. When answering a question, please answer all the subsections of it at once]

1. Define society by mentioning its major characteristics. How is the nature of rural society different from that of urban society? Give your answer by highlighting the important aspects of these two types of society. **12**

2. Is Sociology a science or an art? – Give your arguments in support of your answer. Explain the causes and impacts of the French Revolution? **12**

3. Define social stratification. Discuss the different forms of social stratification with examples in the context of Bangladesh. **12**

4. What is literature review and why is it important in social research? Discuss the different types of social research with advantages and disadvantages. **12**

5. Discuss the different sociological perspectives with examples. Which perspective is more appropriate to analyze the Bangladesh society? - give your arguments in favor of your answer. **12**

6. Why is cybercrime a concern for our society? Identify the reasons for the increasing trend of cybercrime? Give your suggestions for preventing cybercrime. **12**

7. What are the main features of the Fourth Industrial Revolution? How is it affecting social life? Mention the challenges created in Bangladesh as a result of this revolution. **12**